

Self-avoiding walks in Anderson localization

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Contents

Introduction	3
1 The Anderson Model	5
1.1 The Hamiltonian	5
1.2 The spectrum	12
1.3 Main results and Open Problems	14
2 Localization properties	16
2.1 Spectral localization	16
2.2 Dynamical localization	16
3 Localization properties	19
3.1 Spectral localization	19
3.2 Dynamical localization	19
4 Fractional moments	23
4.1 The Green function	23
4.2 Localization at large disorder	23
4.2.1 From fractional moments to dynamical localization . .	25
4.2.2 Exponential decay of the eigenfuctions	33
5 SAW representation	35
5.1 SAW Definitions	35
5.2 SAW representation of G	39
5.3 Proof with SAW	41
6 Connection to Long-Range SAW	43
6.1 The fractional Laplacian	43
6.2 The SAW Representation for Long-Range Resolvents	45
6.3 The Localization Criteria	50

<i>CONTENTS</i>	2
Bibliography	52

Introduction

The Anderson localization model was introduced by the physicist P.W. Anderson in 1958 [And58] to explain the quantum mechanical effects of disorder in materials. In particular, Anderson's purpose was to study the behavior of electrons in a lattice with disordered potential energy on the sites, which is the quantum mechanical analogue of a random walk in a random environment.

The model is governed by a random Schrödinger operator that describes the total energy of the system. This operator is composed of a kinetic term and a random potential term. The kinetic energy allows the particles to hop from site to site, while the random potential disturbs their motion through the material. When the disorder is sufficiently strong, these particles become trapped in a localized region and can no longer spread to conduct electricity. This phenomenon is known as Anderson localization, and it effectively turns the material into an insulator. This stands in sharp contrast to the behavior of electrons in ideal, perfectly ordered crystals, which naturally act as conductors.

In the last decades, the study of random operators has become an important field of mathematical physics. The two important methods that have been used to prove on which condition on the disorder level Anderson localization holds are the *multiscale analysis* (MSA), developed by Fröhlich and Spencer [FS83], and the *fractional moment method* (FMM) by Aizenman and Molchanov [AM93]. The multiscale analysis method focuses on the bound of the second moments of the operator's resolvent, whereas the fractional moments method utilizes its fractional powers. In this thesis, we are going to focus on the second one, since it provides a simple perspective on localization and has enabled exponentially decaying bounds on expectation values of various propagation kernels.

Structure of the thesis

In Chapter 1, we introduce the Anderson model, focusing on the properties of its components, its spectrum, and the main results known to date.

In Chapter 2, we define spectral and dynamical localization. We also define exponential dynamical localization and show how it implies that all moments of the position operator remain bounded in time.

In Chapter 3, we introduce the Green function as the resolvent of the model and explore how the exponential decay of its fractional power leads to localization. Specifically, we also introduce two central analytical tools, such as the *resolvent identity* of linear operators and the resulting *a priori bound*.

In Chapter 4, we provide the preliminary definitions for self-avoiding walk (SAW) methods and obtain the main results of this thesis using a new representation for the Green function.

Finally, Chapter 5 offers a comparison between the standard model with nearest-neighbor jumps and a model with long-range jumps. We define the fractional discrete Laplacian and basically follow the proof structure of the previous Chapters in order to obtain similar results but with a polynomial decay.

Prerequisites

It is useful to have some familiarity with measure theory, basic probabilistic concepts, foundations of the theory of linear operators in Hilbert spaces and the consequences of the spectral theorem for self-adjoint operators.

Chapter 1

The Anderson Model

In this chapter, the random Schrödinger operator is the center of our attention. First, we define it as the *Hamiltonian* and discuss the physical meaning of its time-evolving solutions. Right after, we focus on the properties of each of its components and their spectra. Finally, we discuss the main results and some open problems in the field.

1.1 The Hamiltonian

Transport phenomena in disordered materials are often described via discrete random Schrödinger operators. In quantum mechanics, the *Hamiltonian* represents the total energy of the system, decomposing naturally into kinetic and potential energy. It takes the form of an infinite random matrix on the lattice $\mathbb{Z}^d$,

$$H_\omega = -\Delta + \lambda V_\omega, \quad (1.1.1)$$

where $-\Delta$ is a deterministic matrix representing the kinetic energy, V_ω is a diagonal matrix with random entries representing the disordered potential, and λ is the disorder parameter.

This operator acts on the Hilbert space of square summable sequences

$$\ell^2(\mathbb{Z}^d) = \left\{ u : \mathbb{Z}^d \rightarrow \mathbb{C} \left| \sum_{x \in \mathbb{Z}^d} |u(x)|^2 < \infty \right. \right\}$$

with the inner product defined as $\langle u, v \rangle = \sum_{x \in \mathbb{Z}^d} \overline{u(x)} v(x)$.

The mathematical space $\ell^2(\mathbb{Z}^d)$ is exactly where the physical reality of the electron is recorded. In quantum mechanics, the state of the system is described by a vector in this space : a normalized wave function $\psi \in \ell^2(\mathbb{Z}^d)$, such that

$$\|\psi\|_{\ell^2}^2 = \sum_{x \in \mathbb{Z}^d} |\psi(x)|^2 = 1.$$

The quantity $P(x) = |\psi(x)|^2$ defines a probability mass function on the lattice $\mathbb{Z}^d$, representing the exact probability of finding the particle at site x .

The state functions are the solutions to the time-dependent Schrödinger equation $H_\omega \psi(t) = i\partial_t \psi(t)$. Because the resulting time evolution operator e^{-itH_ω} is unitary on $\ell^2(\mathbb{Z}^d)$, the total probability is conserved at all times $t \in \mathbb{R}$:

$$\|\psi(t)\|_{\ell^2}^2 = \|e^{-itH_\omega} \psi_0\|_{\ell^2}^2 = \|\psi_0\|_{\ell^2}^2 = 1. \quad (1.1.2)$$

In particular, if the particle is initially placed at a specific site $x \in \mathbb{Z}^d$ with $\psi_0 = \delta_x$, the transition probability to find the particle at site $y \in \mathbb{Z}^d$ at time t is given by the squared transition amplitude:

$$P(x \rightarrow y; t) = |\langle \delta_y, e^{-itH_\omega} \delta_x \rangle|^2. \quad (1.1.3)$$

As we will see later, under certain conditions on the disorder parameter, this probability P tends to 0 faster than any polynomial at all times t as the distance $|x - y| \rightarrow \infty$, mathematically proving that the particle remains trapped.

The Discrete Laplacian

The discrete Laplacian is the discrete analogue of the usual Laplacian $(\sum_j \partial^2 / \partial x_j^2)$. For the functions $u \in \ell^2(\mathbb{Z}^d)$ it is defined as

$$-\Delta u(x) := \sum_{y \in \mathbb{Z}^d, |x-y|=1} u(x) - u(y) \quad \text{for } x \in \mathbb{Z}^d. \quad (1.1.4)$$

where $|\cdot|$ denotes the Euclidean norm.

For each site x , there are $2d$ possible y such that $|x - y| = 1$. Thus, we

can define an infinite matrix such that

$$(-\Delta)(x, y) = \begin{cases} 2d & \text{if } x = y \\ -1 & \text{if } |x - y| = 1 \\ 0 & \text{otherwise} \end{cases}$$

The discrete Laplacian represents the kinetic term and connects neighboring sites on the lattice, allowing the electron to transition from one to another.

The operator $-\Delta$ is bounded and self-adjoint on $\ell^2(\mathbb{Z}^d)$. To establish boundedness, we verify that $-\Delta u \in \ell^2(\mathbb{Z}^d)$ for any $u \in \ell^2(\mathbb{Z}^d)$. We can show this through the following,

$$\begin{aligned} \sum_{x \in \mathbb{Z}^d} |-\Delta u(x)|^2 &= \sum_{x \in \mathbb{Z}^d} \left| \sum_{\substack{y \in \mathbb{Z}^d \\ |x-y|=1}} (u(x) - u(y)) \right|^2 \\ &\leq \sum_{x \in \mathbb{Z}^d} \left(2d \sum_{|x-y|=1} |u(x) - u(y)|^2 \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(\sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2 + 2|u(x)u(y)|) \right) \\ &\leq 2d \sum_{x \in \mathbb{Z}^d} \left(2 \sum_{|x-y|=1} (|u(x)|^2 + |u(y)|^2) \right) \\ &= 2d \cdot 2 \cdot 4d \|u\|_{\ell^2}^2 = 16d^2 \|u\|_{\ell^2}^2 < \infty, \end{aligned}$$

where we applied the Cauchy-Schwarz inequality to the sum over the $2d$ neighbors, followed by Young's inequality, and finally the translation invariance of the ℓ^2 norm. This guarantees the operator is bounded:

$$\frac{\|-\Delta u\|_{\ell^2}^2}{\|u\|_{\ell^2}^2} \leq 16d^2 \implies \|-\Delta\| = \sup_{u \neq 0} \frac{\|-\Delta u\|_{\ell^2}}{\|u\|_{\ell^2}} \leq 4d.$$

To prove self-adjointness for a bounded operator, we only need to show

symmetry. For $u, v \in \ell^2(\mathbb{Z}^d)$:

$$\begin{aligned}
\langle u, -\Delta v \rangle &= \sum_{x \in \mathbb{Z}^d} \overline{u(x)} \left(\sum_{|x-y|=1} (v(x) - v(y)) \right) \\
&= \sum_{x \in \mathbb{Z}^d} \left(2d \overline{u(x)} v(x) - \sum_{|x-y|=1} \overline{u(x)} v(y) \right) \\
&= \sum_{x \in \mathbb{Z}^d} 2d \overline{u(x)} v(x) - \sum_{x \in \mathbb{Z}^d} \sum_{|x-y|=1} \overline{u(y)} v(x) \\
&= \sum_{x \in \mathbb{Z}^d} v(x) \left(\sum_{|x-y|=1} (\overline{u(x)} - \overline{u(y)}) \right) \\
&= \sum_{x \in \mathbb{Z}^d} v(x) \overline{(-\Delta u(x))} = \langle -\Delta u, v \rangle.
\end{aligned}$$

The spectrum of a bounded, self-adjoint operator is a compact subset of $\mathbb{R}$, which in this case is exactly the closed interval $[0, 4d]$. This range represents the allowed kinetic energy levels an electron can possess.

To understand this result explicitly, we apply the discrete Fourier transform:

$$\mathcal{F}: \ell^2(\mathbb{Z}^d) \longrightarrow L^2([0, 2\pi)^d)$$

defined by

$$(\mathcal{F}u)(k) := \lim_{N \rightarrow \infty} (2\pi)^{-\frac{d}{2}} \sum_{|x| \leq N} u(x) e^{-ik \cdot x}, \quad k \in [0, 2\pi)^d.$$

We first assume $u \in \ell^1(\mathbb{Z}^d)$. Because $\ell^1(\mathbb{Z}^d)$ is a dense subset of $\ell^2(\mathbb{Z}^d)$ and the Fourier transform $\mathcal{F}$ is unitary, we can extend the results by con-

tinuity to all functions in $\ell^2(\mathbb{Z}^d)$.

$$\begin{aligned}
(\mathcal{F}(-\Delta u))(k) &= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} (-\Delta u)(x) e^{-ik \cdot x} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(\sum_{|y-x|=1} (u(x) - u(y)) \right) e^{-ik \cdot x} \\
&= (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \left(2d u(x) - \sum_{|y-x|=1} u(y) \right) e^{-ik \cdot x} \\
&= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{x \in \mathbb{Z}^d} \sum_{|y-x|=1} u(y) e^{-ik \cdot x} \\
&= 2d (\mathcal{F}u)(k) - (2\pi)^{-\frac{d}{2}} \sum_{y \in \mathbb{Z}^d} u(y) \sum_{|m|=1} e^{-ik \cdot (y+m)} \quad (\text{setting } m = x - y) \\
&= 2d (\mathcal{F}u)(k) - \left((2\pi)^{-\frac{d}{2}} \sum_{y \in \mathbb{Z}^d} u(y) e^{-ik \cdot y} \right) \sum_{|m|=1} e^{-ik \cdot m} \\
&= 2d (\mathcal{F}u)(k) - (\mathcal{F}u)(k) \sum_{j=1}^d (e^{-ik_j} + e^{ik_j}) \\
&= \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) (\mathcal{F}u)(k).
\end{aligned}$$

Thus, the Laplacian acts as a simple multiplication operator in Fourier space

$$\mathcal{F}(-\Delta)\mathcal{F}^{-1} = 2 \sum_{j=1}^d (1 - \cos(k_j))$$

on $L^2([0, 2\pi)^d)$ in the momentum variable $k = (k_1, \dots, k_d)$.

Remark 1.1.1. The discrete Laplacian has no true eigenvalues in $\ell^2(\mathbb{Z}^d)$. However, similar to the continuum Laplacian, it possesses plane waves as generalized eigenfunctions. For a fixed momentum $k \in [0, 2\pi)^d$, we define:

$$\phi_k(x) := e^{-ix \cdot k}.$$

While $\phi_k \notin \ell^2(\mathbb{Z}^d)$ because it is not square-summable, the operator $(-\Delta)$

acts on it as

$$\begin{aligned}
(-\Delta\phi_k)(x) &= \sum_{|m|=1} (e^{-ix \cdot k} - e^{-i(x+m) \cdot k}) \\
&= 2d\phi_k(x) - \left(\sum_{j=1}^d 2\cos(k_j) \right) \phi_k(x) \\
&= \left(2 \sum_{j=1}^d (1 - \cos(k_j)) \right) \phi_k(x).
\end{aligned}$$

Thus, ϕ_k is a bounded, generalized eigenfunction of $(-\Delta)$ corresponding to the spectral value $2 \sum_{j=1}^d (1 - \cos(k_j))$. Since $\cos(k_j)$ oscillates between -1 and 1 , the spectrum of the discrete Laplacian is purely continuous and strictly bounded within the interval $[0, 4d]$.

The Random Potential

The random potential is expressed as the multiplication operator $V_\omega(x) = \omega_x$. It is a diagonal matrix, where $\omega = (\omega_x)_{x \in \mathbb{Z}^d}$ is a set of independent identically distributed real-valued random entries. Physically, V_ω represents the random electric potential created by nuclei at the sites $x \in \mathbb{Z}^d$.

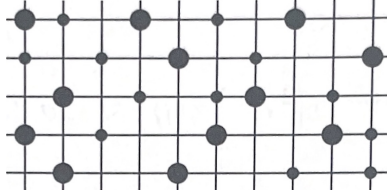


Figure 1.1: A disordered lattice system. Picture from [AW15b]

Definition 1.1.2 (i.i.d. random variables). Let $(\Omega, \mathcal{A}, \mathbb{P})$ be a general probability space. A family of random variables $(\omega_x)_{x \in \mathbb{Z}^d}$ defined on $(\Omega, \mathcal{A}, \mathbb{P})$ is **identically distributed** if there exists a Borel probability measure μ on $\mathbb{R}$ such that $\mathbb{P}(\omega_x \in A) = \mu(A)$ for all $x \in \mathbb{Z}^d$ and all Borel sets $A \subset \mathbb{R}$. The family $(\omega_x)_{x \in \mathbb{Z}^d}$ is **independent** if

$$\mathbb{P}(\omega_{x_1} \in A_1, \dots, \omega_{x_l} \in A_l) = \prod_{j=1}^l \mathbb{P}(\omega_{x_j} \in A_j) = \prod_{j=1}^l \mu(A_j)$$

for every finite subset of distinct indices $\{x_1, \dots, x_l\} \subset \mathbb{Z}^d$ and arbitrary

Borel sets $A_1, \dots, A_l \subset \mathbb{R}$.

In this thesis we assume that ω_x are uniformly distributed in the real closed interval $[-1, 1]$.

We consider then the probability space as the following infinite product space

$$(\Omega, \mathcal{A}, \mathbb{P}) = \bigotimes_{x \in \mathbb{Z}^d} ([-1, 1], \mathcal{B}([-1, 1]), \mu),$$

where $\mathcal{B}([-1, 1])$ is the Borel σ -algebra and $d\mu(\omega_x) = \rho(\omega_x)d\omega_x$ where the continuous density is given by $\rho = \frac{1}{2}\chi_{[-1, 1]}$.

Remark 1.1.3. It is possible to generalise the results to all absolutely continuous distributions where ρ has compact support.

As a diagonal operator, the spectrum of V_ω consists of the random entries ω_x , each corresponding to an eigenvector given by the Dirac delta $\delta_x \in \ell^2(\mathbb{Z}^d)$.

As for the discrete laplacian, the random term is summable in $\ell^2(\mathbb{Z}^d)$, bounded and self-adjoint. Self-adjointness of the maximal multiplication operator by a real-valued function is a fact. For definition V_ω is almost surely bounded in the closed interval $[-1, 1] \subset \mathbb{R}$, and since $\|V_\omega u\|_{\ell^2}^2 \leq \|u\|_{\ell^2}^2$ it is also summable in $\ell^2(\mathbb{Z}^d)$.

The Disorder Parameter

The disorder parameter is represented by $\lambda > 0$. The value of λ is a great indicator for localization, which makes it one of the central objects of interest of this thesis. It allows us to adjust the strength of the random disorder with respect to the deterministic part.

Let's consider the two extreme cases:

- For $\lambda = 0$ the Hamiltonian loses the random term and only considers the Laplacian. Without disorder, the material behaves like a perfect crystal without localization. The state functions are continuous plane waves and the spectrum is also continuous between $[0, 4d]$.
- For $\lambda \gg 1$ the kinetic hopping term becomes negligible compared to the random potential, so that $H_\omega \approx \lambda V_\omega$. In this scenario electrons are unable to hop between adjacent sites, and the material behaves as an insulator. In this limit, the system exhibits trivial localization:

the eigenfunctions are localized Dirac deltas δ_x , and the almost sure spectrum is confined to the compact interval $[-\lambda, \lambda]$

1.2 The spectrum

The next goal is to determine the spectrum of H_ω . Because the random potential entries $(\omega_x)_{x \in \mathbb{Z}^d}$ are independent and identically distributed, the random Hamiltonian $H_\omega = -\Delta + \lambda V_\omega$ forms an ergodic family of bounded self-adjoint operators. A standard consequence of ergodic operator theory is that the spectrum of H_ω is almost surely deterministic: there exists a closed, non-random set $\Sigma \subset \mathbb{R}$ such that

$$\sigma(H_\omega) = \Sigma \quad \text{for } \mathbb{P}\text{-almost all } \omega \in \Omega. \quad (1.2.1)$$

More specifically, as proven in [Sto11, Theorem 2], this almost sure spectrum can be determined explicitly as the Minkowski sum of the kinetic energy spectrum and the support of the random potential distribution:

$$\Sigma = \sigma(-\Delta) + \lambda \operatorname{supp}(\mu). \quad (1.2.2)$$

Recall from Section 1.1 that the spectrum of the discrete Laplacian is the band $\sigma(-\Delta) = [0, 4d]$. Since the random potential V_ω has i.i.d. entries distributed uniformly on $[-1, 1]$, the almost sure spectrum of λV_ω is the interval $[-\lambda, \lambda]$. Consequently, the almost sure spectrum of H_ω is given by the compact interval:

$$\sigma(H_\omega) = [0, 4d] + [-\lambda, \lambda] = [-\lambda, 4d + \lambda]. \quad (1.2.3)$$

This confirms that the spectrum is a strictly bounded subset of $\mathbb{R}$.

Preliminaries on Spectral Measures

Before characterizing the physical meaning of this spectrum, we briefly recall the relevant spectral theory. Let μ be a Borel measure on $\mathbb{R}$. Consider the set of **pure points** of μ , defined as $P = \{x \in \mathbb{R} \mid \mu(\{x\}) \neq 0\}$. We define the pure point measure as

$$\mu_{\text{pp}}(X) = \mu(P \cap X)$$

for any Borel set $X \subset \mathbb{R}$. By definition, there remains a continuous measure μ_c such that $\mu_c(\{x\}) = 0$ for all $x \in \mathbb{R}$, satisfying $\mu = \mu_{\text{pp}} + \mu_c$. Because $\mu_{\text{pp}}(P^c) = 0$ and $\mu_c(P) = 0$, these two measures are concentrated on disjoint sets and are therefore mutually singular.

This decomposition naturally extends to spectral measures:

Definition 1.2.1. Let $A : \mathcal{H} \rightarrow \mathcal{H}$ be a bounded, self-adjoint operator. For any vector $\phi \in \mathcal{H}$ and any continuous function $f \in C(\sigma(A))$, there exists a unique Borel measure μ_ϕ on the compact set $\sigma(A)$ satisfying

$$\langle \phi, f(A)\phi \rangle = \int_{\sigma(A)} f(\xi) d\mu_\phi(\xi).$$

The measure μ_ϕ is called the *spectral measure associated with the vector ϕ* .

By the spectral theorem for self-adjoint operators (see [Rud87] and [RS80]), the Hilbert space (such as $\ell^2(\mathbb{Z}^d)$) decomposes uniquely into an orthogonal direct sum of invariant subspaces:

$$\mathcal{H} = \mathcal{H}_{\text{pp}} \oplus \mathcal{H}_c, \quad (1.2.4)$$

where $\mathcal{H}_{\text{pp}} = \{\phi \in \mathcal{H} \mid \mu_\phi \text{ is pure point}\}$ and $\mathcal{H}_c = \{\phi \in \mathcal{H} \mid \mu_\phi \text{ is continuous}\}$. For bounded self-adjoint operators, these subspaces define two (not necessarily disjoint) subsets of the spectrum:

$$\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega) \cup \sigma_c(H_\omega), \quad (1.2.5)$$

where $\sigma_{\text{pp}}(H_\omega) = \overline{\{\lambda \in \mathbb{R} \mid \lambda \text{ is an eigenvalue of } H_\omega\}}$.

Remark 1.2.2. In measure theory, the continuous measure μ_c further decomposes uniquely into $\mu_{\text{ac}} + \mu_{\text{sc}}$, where μ_{ac} is absolutely continuous with respect to Lebesgue measure, and μ_{sc} is singular continuous. Following this principle, the continuous subspace can also be decomposed as $\mathcal{H}_c = \mathcal{H}_{\text{ac}} \oplus \mathcal{H}_{\text{sc}}$.

Physical Interpretation and the RAGE Theorem

From a physical perspective, states in $\mathcal{H}_{\text{pp}}$ correspond to bound states that remain confined to finite regions for all time, while states in the continuous subspace $\mathcal{H}_c$ represent scattering states associated with transport and wave-packet spreading. This heuristic meaning is made mathematically rigorous by the RAGE theorem (named after Ruelle, Amrein, Georgescu, and Enss):

Theorem 1.2.3 (RAGE Theorem). *Let H be a self-adjoint Schrödinger operator on $\ell^2(\mathbb{Z}^d)$. Then:*

(i) $\phi \in \mathcal{H}_{\text{pp}}$ if and only if

$$\lim_{R \rightarrow \infty} \sup_{t \in \mathbb{R}} \|\chi_{\{|x| > R\}} e^{-itH} \phi\| = 0. \quad (1.2.6)$$

(ii) $\psi \in \mathcal{H}_c$ if and only if for all $R > 0$,

$$\lim_{T \rightarrow \infty} \frac{1}{2T} \int_{-T}^T \|\chi_{\{|x| \leq R\}} e^{-itH} \psi\|^2 dt = 0. \quad (1.2.7)$$

Here, $\chi_{\{|x| \leq R\}}$ and $\chi_{\{|x| > R\}}$ denote the projection operators corresponding to multiplication by the characteristic functions of the ball $\{x \in \mathbb{Z}^d \mid |x| \leq R\}$ and its complement, respectively.

In terms of this classification, Anderson localization corresponds exactly to the absence of continuous spectrum, ensuring that $\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega)$ almost surely.

1.3 Main results and Open Problems

There are many studies on the Anderson localization model by both physicists and mathematicians. Various discoveries have been made over the years, yet deep challenges remain.

The main result of this thesis states that if $s \in (0, 1)$ and $\lambda > \lambda_0(s)$,

$$\mathbb{E} (|(H_\omega - z)^{-1}(x, y)|^s) \leq C e^{-\mu|x-y|} \quad (1.3.1)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and all $x, y \in \mathbb{Z}^d$, with $C < \infty$ and $\mu > 0$. Here, $\lambda_0 = \lambda_{\text{And}}$ is a fixed threshold proposed by P.W. Anderson ([And58]), which was rigorously proven later by J. Schenker ([Sch15]) using self-avoiding walks.

Naturally, the resolvent is well-defined only when $z \notin \sigma(H_\omega)$. The *fractional moments method* was revolutionary because averaging a fractional power of the resolvent avoids these divergences, allowing the bound to hold uniformly for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Originally, the absence of conductivity was proven using integer powers instead of fractional ones. This creates divergence problems near zero and requires complicated analytical tools like the MSA method.

The intuition behind the fractional method is simple and can be understood from the following integral, which we will encounter in upcoming proofs (for example, see Lemma 4.2.3):

$$\int_{\Sigma} \frac{1}{|x|^s} dx \begin{cases} < \infty & \text{for } s \in (0, 1), \\ = \infty & \text{for } s \geq 1 \text{ (diverges at } x = 0), \end{cases}$$

where $\Sigma \subset \mathbb{R}$ is a closed interval containing the origin.

Very important results concern the value of λ . Based on the dimension of the grid, we know the following.

- $d = 1$.

For the one-dimensional grid, it has been proven, that localization holds for all $\lambda > 0$ [GMP77].

- $d = 2$.

For $d = 2$ it has been numerically established that localization persists at arbitrarily small disorder.

- $d \geq 3$.

For $d \geq 3$, to find the exact value of λ , until which the material is still a conductor, is an open problem. The best that we can prove is given in this thesis and only states, that localization shows from a specific λ_0 on (see definition in Theorem 4.2.1).

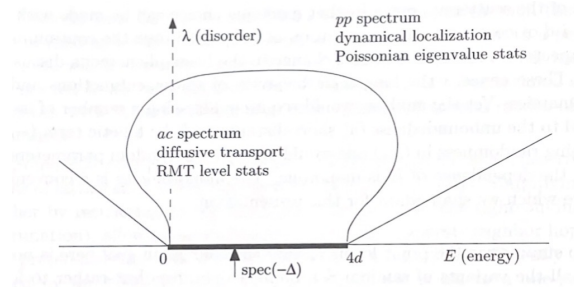


Figure 1.2: Picture from [AW15b]

Chapter 2

Localization properties

We understood what localization means from an heuristic point of view. In the previous Chapter, it came out as a spectral property of the Hemiltonian. In this section, we also introduce dynamical properties for localization, together with the implications between them. The definitions follow the ones given by A. Klein in [Kle07].

2.1 Spectral localization

Definition 2.1.1 (Spectral localization). We say that $H = H_\omega$ exhibits **spectral localization** if it has pure point spectrum almost surely. That is

$$\sigma(H_\omega) = \sigma_{pp}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad (2.1.1)$$

for almost every realization of $\omega \in \Omega$.

2.2 Dynamical localization

Definition 2.2.1 (Dynamical localization). We say that H_ω shows dynamical localization if for all $p > 0$ and $\phi \in \ell^2(\mathbb{Z}^d)$,

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \phi \| < \infty \quad (2.2.1)$$

for almost every realization of $\omega \in \Omega$, where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$.

Definition 2.2.2 (Exponential localization). We say that H_ω shows expo-

nential localization if

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |e^{-itH_\omega}(0, x)| \right) \leq C e^{-\mu|x|} \quad (2.2.2)$$

for $x \in \mathbb{Z}^d$

Furthermore the eigenfunctions corresponding to all eigenvalues E in the almost sure spectrum, decay exponentially: for almost all ω , H_ω has a complete set of eigenfunctions $\phi_E(\cdot, \omega)$ satisfying

$$|\phi_E(x, \omega)| \leq C_\omega e^{-\mu|x-x_E(\omega)|}$$

where $\mu > 0$ is the localization length parameter, $x_E(\omega)$ are the centers of localization, and $C_\omega < \infty$ is a finite constant.

Remark 2.2.3. Dynamical localization implies that all moments of the distance between the two sites are bounded uniformly in time (recall that we are working in an infinite grid, and with translation invariance, so as well as $(0, x)$ we can consider any site in the grid). From (3.2.1) we already notice that the probability of transitioning from 0 to x goes to zero faster than any polynomial, but (3.2.2) explicits this directly with an exponential bound. In fact, the two definitions are strongly related and we can prove the following implication.

Proposition 2.2.4. *Exponential localization implies that all moments of the position operator are bounded in time for almost every realisation of V_ω , i.e., dynamical localisation.*

Proof. To see how (2.1) follows from (2.2), we assume that $\psi(k) = 0$ for $|k| > R$, since ψ has compact support. Then

$$\begin{aligned} \| |X|^p e^{-itH_\omega} \psi \|^2 &= \sum_j |\langle \delta_j, |X|^p e^{-itH_\omega} \psi \rangle|^2 \\ &= \sum_j |j|^{2p} \left| \sum_{|k| \leq R} e^{-itH_\omega}(j, k) \psi(k) \right|^2 \\ &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} |e^{-itH_\omega}(j, k)|^2 \|\psi\|^2. \end{aligned}$$

where the last step used the Cauchy-Schwarz inequality.

Since $|e^{-itH_\omega}(j, k)|^2 \leq 1$:

$$\begin{aligned} \mathbb{E}(\sup_t ||X|^p e^{-itH_\omega} \psi||^2) &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} \mathbb{E}(\sup_t |e^{-itH_\omega}(j, k)|) ||\psi||^2 \\ &\leq C \sum_j \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} ||\psi||^2 \\ &< \infty. \end{aligned}$$

□

Remark 2.2.5. Dynamical localisation holds *almost surely* rather than for every realisation, because our bounds control only the *expected* value over the disorder. A bound on the average does not exclude that a single realisation is large. However, if the expectations decay fast enough, then being large at distance $|x|$ is a very unlikely event for each x , and the probabilities of these events are summable. The Borel–Cantelli lemma below then shows that, with probability one, this can happen only for finitely many x .

Lemma 2.2.6 (Borel–Cantelli). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ a probability space. Let $n \rightarrow A_n \in \mathcal{F}$ a sequence of events. Then if $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$, almost surely, only finitely many of the A_n occur, which means $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$*

Example 2.2.7 (From proposition 3.2.4). Set $X_x := \sup_{t \in \mathbb{R}} |e^{-itH_\omega}(0, x)|$ and consider the events

$$A_x := \{X_x > e^{-\mu|x|/2}\}.$$

By Markov’s inequality and (3.2.2),

$$\mathbb{P}(A_x) \leq \frac{\mathbb{E}[X_x]}{e^{-\mu|x|/2}} \leq C e^{-\mu|x|/2}.$$

This is summable over $x \in \mathbb{Z}^d$, so $\sum_x \mathbb{P}(A_x) < \infty$. By Lemma 3.2.6, almost surely only finitely many A_x occur. Hence there is a random radius $R(\omega) < \infty$ such that $X_x \leq e^{-\mu|x|/2}$ for all $|x| > R(\omega)$.

Split the sum at $R(\omega)$, considering $X_x \leq 1$:

$$\sum_j \sum_{|k| \leq R} |j|^{2p} |e^{-itH_\omega}(j, k)|^2 ||\psi||^2 \leq \sum_{|j| \leq R(\omega)} |j|^{2p} ||\psi||^2 + \sum_{|j| > R(\omega)} \sum_{|k| \leq R} |j|^p e^{-\mu|j-k|} ||\psi||^2 < \infty.$$

for almost every realization of $\omega \in \Omega$.

Chapter 3

Localization properties

We have understood what localization means from a heuristic point of view. In the previous chapter, it was presented as a spectral property of the Hamiltonian. In this section, we introduce dynamical properties for localization and discuss the implications between them. The definitions follow those given by A. Klein in [Kle07].

3.1 Spectral localization

Definition 3.1.1 (Spectral localization). We say that $H = H_\omega$ exhibits **spectral localization** if it has pure point spectrum almost surely. That is,

$$\sigma(H_\omega) = \sigma_{\text{pp}}(H_\omega) \quad \text{and} \quad \sigma_c(H_\omega) = \emptyset \quad (3.1.1)$$

for almost every realization of $\omega \in \Omega$.

3.2 Dynamical localization

Definition 3.2.1 (Dynamical localization). We say that H_ω exhibits dynamical localization if for all $p > 0$ and $\phi \in \ell^2(\mathbb{Z}^d)$ of finite support,

$$\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \phi \| < \infty \quad (3.2.1)$$

for almost every realization of $\omega \in \Omega$, where the position operator $|X|$ is defined by $(|X|\phi)(n) = |n|\phi(n)$.

Definition 3.2.2 (Exponential localization). We say that H_ω exhibits ex-

ponential localization if

$$\mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle \delta_0, e^{-itH_\omega} \delta_x \rangle| \right) \leq C e^{-\mu|x|} \quad (3.2.2)$$

for all $x \in \mathbb{Z}^d$.

Furthermore, the eigenfunctions corresponding to all eigenvalues E in the almost sure spectrum decay exponentially: for almost every ω , H_ω has a complete set of eigenfunctions $\phi_E(\cdot, \omega)$ satisfying

$$|\phi_E(x, \omega)| \leq C_\omega e^{-\mu|x - x_E(\omega)|}$$

where $\mu > 0$ is the localization length parameter, $x_E(\omega)$ are the centers of localization, and $C_\omega < \infty$ is a finite constant.

Remark 3.2.3. Dynamical localization implies that all spatial moments of the distance are bounded uniformly in time. (Recall that because we are working on an infinite grid with translation invariance, we can consider any arbitrary starting site just as easily as the origin 0).

From (3.2.1), we can already deduce that the probability of transitioning from 0 to x goes to zero faster than any polynomial, but (3.2.2) makes this explicit with a direct exponential bound. In fact, the two definitions are strongly related, and we can prove the following implication.

Proposition 3.2.4. *Exponential localization implies that all moments of the position operator are bounded in time for almost every realization of V_ω , i.e., it implies dynamical localization.*

Proof. To see how (3.2.1) follows from (3.2.2), assume that $\psi(k) = 0$ for $|k| > R$ (i.e., ψ has finite support). Then

$$\begin{aligned} \| |X|^p e^{-itH_\omega} \psi \|^2 &= \sum_j |\langle \delta_j, |X|^p e^{-itH_\omega} \psi \rangle|^2 \\ &= \sum_j |j|^{2p} \left| \sum_{|k| \leq R} \langle \delta_j, e^{-itH_\omega} \delta_k \rangle \psi(k) \right|^2 \\ &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} |\langle \delta_j, e^{-itH_\omega} \delta_k \rangle|^2 \|\psi\|^2, \end{aligned}$$

where the last step applies the Cauchy-Schwarz inequality.

Because $|\langle \delta_j, e^{-itH_\omega} \delta_k \rangle| \leq 1$, it follows

$$\begin{aligned} \mathbb{E} \left(\sup_{t \in \mathbb{R}} \| |X|^p e^{-itH_\omega} \psi \|^2 \right) &\leq \sum_j \sum_{|k| \leq R} |j|^{2p} \mathbb{E} \left(\sup_{t \in \mathbb{R}} |\langle \delta_j, e^{-itH_\omega} \delta_k \rangle| \right) \|\psi\|^2 \\ &\leq C \sum_j \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 \\ &< \infty. \end{aligned} \tag{3.2.3}$$

□

Remark 3.2.5. Dynamical localization holds *almost surely* rather than for every single realization because our bounds control only the *expected* value over the disorder. A bound on the average does not strictly exclude the possibility that a specific realization is large. However, if the expectations decay fast enough, then being large at a great distance $|x|$ is a very unlikely event. The probabilities of these events are summable, and the Borel–Cantelli lemma below guarantees that, with probability one, this can happen only for finitely many x .

Lemma 3.2.6 (Borel–Cantelli). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(A_n)_{n \in \mathbb{N}}$ be a sequence of events in $\mathcal{F}$. If $\sum_{n \in \mathbb{N}} \mathbb{P}(A_n) < \infty$, then almost surely, only finitely many of the events A_n occur. That is, $\mathbb{P}(\limsup_{n \rightarrow \infty} A_n) = 0$.*

Example 3.2.7 (From Proposition 3.2.4). Set $X_x := \sup_{t \in \mathbb{R}} |\langle \delta_0, e^{-itH_\omega} \delta_x \rangle|$ and consider the events

$$A_x := \{X_x > e^{-\mu|x|/2}\}.$$

By Markov’s inequality and (3.2.2),

$$\mathbb{P}(A_x) \leq \frac{\mathbb{E}[X_x]}{e^{-\mu|x|/2}} \leq C e^{-\mu|x|/2}.$$

This is summable over $x \in \mathbb{Z}^d$, meaning $\sum_x \mathbb{P}(A_x) < \infty$.

By Lemma 3.2.6, almost surely only finitely many A_x occur. Hence, there exists a random, finite radius $R(\omega) < \infty$ such that $X_x \leq e^{-\mu|x|/2}$ for all $|x| > R(\omega)$.

We can then split the sum at $R(\omega)$ and use the trivial bound $X_x \leq 1$ for

the interior terms:

$$\sum_j \sum_{|k| \leq R} |j|^{2p} X_{j-k}^2 \|\psi\|^2 \leq \sum_{|j| \leq R(\omega)} \sum_{|k| \leq R} |j|^{2p} \|\psi\|^2 + \sum_{|j| > R(\omega)} \sum_{|k| \leq R} |j|^{2p} e^{-\mu|j-k|} \|\psi\|^2 < \infty$$

for almost every realization of $\omega \in \Omega$.

Chapter 4

Fractional moments

In this section, we introduce the key lemmas and theorems that help us prove localization. The goal of this section is to demonstrate the power of the fractional moments method (FMM). First, we define the Green function, and then we prove the first fractional bound: the *a priori bound*. This will provide a good example of how we use simple but important analytic tools, such as the *resolvent identity*, to obtain strong results. Finally, we will prove dynamical and spectral localization using the bound on the fractional power of the Green function.

4.1 The Green function

The *resolvent operator* is defined as:

$$G_z := (H_\omega - z)^{-1} \tag{4.1.1}$$

and is a well-defined bounded operator for all $z \in \mathbb{C} \setminus \mathbb{R}$.

4.2 Localization at large disorder

The primary goal of this thesis is to prove the following theorem.

Theorem 4.2.1 (Aizenman–Molchanov and Schenker). *Let $0 < s < 1$. Then there exists $\lambda_0 > 0$ such that for $\lambda > \lambda_0$, there are constants $C < \infty$ and $\mu > 0$ satisfying*

$$\mathbb{E}(|G_z(x, y)|^s) \leq C e^{-\mu|x-y|} \tag{4.2.1}$$

uniformly in $x, y \in \mathbb{Z}^d$ and $z \in \mathbb{C} \setminus \mathbb{R}$. In particular, λ_0 is the unique solution to the equation $\lambda_0 = \mu_d e \ln(\lambda_0)$, and for all $\lambda > \lambda_0 > e$ and $\epsilon > 0$ we have

$$\mathbb{E} \left(|G_z(x, y)|^{1 - \frac{1}{\ln(\lambda)}} \right) \leq \beta(\lambda) K_\epsilon \ln(\lambda) e^{-m_\epsilon(\lambda)|x-y|} \quad (4.2.2)$$

where

$$m_\epsilon(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

Remark 4.2.2. We distinguish (4.2.1) from (4.2.2) in this theorem because the first represents a generalized bound for any absolutely continuous distribution of ω , while the second considers the specific case of a uniform distribution on $[-1, 1]$, yielding a more precise value for λ_0 . Both are equally valid for proving localization, but our ultimate goal is to reach (4.2.2).

Before achieving this result, we introduce two preparatory lemmas: the so-called *a priori bound* on fractional moments and the *resolvent identity*.

Lemma 4.2.3 (A priori bound). *Let $s \in (0, 1)$. Then it holds*

$$\mathbb{E}_x(|G_z(x, x)|^s) \leq \frac{1}{(1-s)\lambda^s} \quad (4.2.3)$$

for all $x \in \mathbb{Z}^d$, $z \in \mathbb{C} \setminus \mathbb{R}$, and $\lambda > 0$.

Lemma 4.2.4 (Resolvent identity). *For a linear operator $A : \mathcal{D}(A) \rightarrow \mathcal{H}$ and $z_1, z_2 \in \varrho(A)$,*

$$(A - z_1)^{-1} - (A - z_2)^{-1} = (A - z_1)^{-1}(z_1 - z_2)(A - z_2)^{-1}.$$

Moreover, if $B : \mathcal{D}(B) \rightarrow \mathcal{H}$ shares a common domain with A such that $\mathcal{D}(A) = \mathcal{D}(B)$, then

$$(A - z)^{-1} - (B - z)^{-1} = (A - z)^{-1}(B - A)(B - z)^{-1}$$

for any $z \in \varrho(A) \cap \varrho(B)$.

The resolvent identity follows directly from simple algebraic manipulation of the inverses.

Proof of the a priori bound. This proof demonstrates the simplicity of the fundamental idea underlying the FMM. For a fixed $x \in \mathbb{Z}^d$, we write $\omega = (\hat{\omega}, \omega_x)$, where $\hat{\omega}$ is shorthand for the variables at all other sites, $(\omega_n)_{n \in \mathbb{Z}^d \setminus \{x\}}$. Using $P_{e_x} := \langle e_x, \cdot \rangle e_x$ to denote the orthogonal projection onto the span of

the basis vector e_x , we can separate the dependence of H_ω on ω_x and $\hat{\omega}$ as follows :

$$H_\omega = -\Delta + \lambda V_{\hat{\omega}} + \lambda \omega_x P_{e_x} = H_{\hat{\omega}} + \lambda \omega_x P_{e_x}.$$

For the operators $(H_\omega - z)$ and $(H_{\hat{\omega}} - z)$, the resolvent identity yields

$$(H_\omega - z)^{-1} = (H_{\hat{\omega}} - z)^{-1} - \lambda \omega_x (H_{\hat{\omega}} - z)^{-1} P_{e_x} (H_\omega - z)^{-1}. \quad (4.2.4)$$

Taking the matrix elements, it follows

$$G_z(x, x) = \hat{G}_z(x, x) - \lambda \omega_x \hat{G}_z(x, x) G_z(x, x), \quad (4.2.5)$$

where $\hat{G}_z$ denotes the Green function of $H_{\hat{\omega}}$. Solving for $G_z(x, x)$, we obtain

$$G_z(x, x) = \frac{1}{a + \lambda \omega_x} \quad \text{with } a = \frac{1}{\hat{G}_z(x, x)}. \quad (4.2.6)$$

Note that this expression is well-defined because one can easily check through the resolvent identity $\text{Im} \hat{G}_z(x, x) > 0$ whenever $\text{Im} z > 0$.

The crucial fact is that the complex number a does not depend on ω_x . Thus, defining the partial expectation over the single site x as $\mathbb{E}_x(\dots) := \int \dots \rho(\omega_x) d\omega_x$, and using the uniform density $\rho(\omega_x) = \frac{1}{2} \chi_{[-1, 1]}$, we find

$$\mathbb{E}_x(|G_z(x, x)|^s) = \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{\left|\frac{a}{\lambda} + \omega_x\right|^s} \leq \frac{1}{2\lambda^s} \int_{-1}^1 \frac{d\omega_x}{|\omega_x|^s} = \frac{1}{(1-s)\lambda^s}. \quad (4.2.7)$$

The resulting constant $\frac{1}{1-s}$ is independent of λ and a , and therefore independent of the remaining random variables $\hat{\omega}$, the energy z , and the site x . $\square$

4.2.1 From fractional moments to dynamical localization

In this subsection we aim to prove the following theorem:

Theorem 4.2.5. *If there exist $s \in (0, 1)$, $C < \infty$ and $\mu > 0$ such that*

$$\mathbb{E}(|G_{E+i\eta}(x, y)|^s) \leq C e^{-\mu|x-y|} \quad (4.2.8)$$

uniformly in $E \in \mathbb{R}$ and $\eta > 0$, then exponential localization holds.

Before proving this theorem, we need two more lemmas.

Lemma 4.2.6. *For a bounded and self-adjoint operator A ,*

$$\|(A - z)^{-1}\| \leq \frac{1}{\text{Im}(z)} \quad (4.2.9)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$.

Proof. For the definition of the operator norm, we know that

$$\|(A - z)^{-1}\| := \sup_{\mu \in \sigma((A - z)^{-1})} |\mu|$$

Let's consider a $\lambda \in \mathbb{C}$ and write $\mu = \frac{1}{\lambda - z}$. This means:

$$\begin{aligned} \frac{1}{\lambda - z} \in \sigma((A - z)^{-1}) &\iff (A - z)^{-1} - (\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - z)^{-1}(A - \lambda)(\lambda - z)^{-1} \text{ not invertible} \\ &\iff (A - \lambda) \text{ not invertible} \\ &\iff \lambda \in \sigma(A) \end{aligned}$$

Where the resolvent identity has been used in third Implication.

Finally we obtain:

$$\|(A - z)^{-1}\| = \sup_{\lambda \in \sigma(A)} \frac{1}{|\lambda - z|} = \frac{1}{\inf_{\lambda \in \sigma(A)} |\lambda - z|} = \frac{1}{\text{dist}(\sigma(A), z)}$$

For a bounded and self-adjoint operator, we know that the spectrum is real-valued. For $z \in \mathbb{C} \setminus \mathbb{R}$ the minimum distance between z and $\sigma(A)$ is the imaginary component of z :

$$\|(A - z)^{-1}\| \leq \frac{1}{|\text{Im}(z)|} \quad (4.2.10)$$

□

The next proposition shows one of the most important correlation between the fractional and second moments.

Proposition 4.2.7. *For every $s \in (0, 1)$ there exists a constant $C_2 := C_2(s, \rho) < \infty$ such that*

$$|\text{Im } z| \mathbb{E}_x (|G_z(x, y)|^2) \leq C_2 \mathbb{E}_x (|G_z(x, y)|^s) \quad (4.2.11)$$

for all $z \in \mathbb{C} \setminus \mathbb{R}$ and $x, y \in \mathbb{Z}^d$.

Proof. As in the proof of 4.2.3 we consider the Hamiltonian

$$H^{(\alpha)} = H_\omega + \alpha P_{e_x}.$$

Its Green function will be denoted by $G^{(\alpha)}$. Applying the resolvent identity, we find

$$(H_\omega - z)^{-1} = (H^{(\alpha)} - z)^{-1} + \alpha(H_\omega - z)^{-1} P_{e_x} (H^{(\alpha)} - z)^{-1}$$

and

$$\begin{aligned} G_z(x, y) &= G_z^{(\alpha)}(x, y) + \alpha G_z(x, y) G^{(\alpha)}(x, x) \\ \Rightarrow G_z^{(\alpha)}(x, y) &= \frac{G_z(x, y)}{1 + \alpha G_z(x, x)} \\ &= \frac{1}{\alpha + G_z(x, x)^{-1}} \cdot \frac{G_z(x, y)}{G_z(x, x)}. \end{aligned} \quad (4.2.12)$$

For the special case $x = y$ and $\tilde{\alpha} = -\operatorname{Re} G_z(x, x)^{-1}$ we get from (4.2.12) and Lemma 4.2.6 that

$$\left| \frac{1}{\operatorname{Im} G_z(x, x)^{-1}} \right| = |G_z^{(\tilde{\alpha})}(x, x)| \leq \frac{1}{|\operatorname{Im} z|},$$

i.e. $|\operatorname{Im} G_z(x, x)^{-1}| \geq |\operatorname{Im} z|$. Inserting this into (4.2.12) gives

$$|\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 \leq \frac{|\operatorname{Im} G_z(x, x)^{-1}|}{|\alpha + G_z(x, x)^{-1}|^2} \cdot \frac{|G_z(x, y)|^2}{|G_z(x, x)|^2}. \quad (4.2.13)$$

On the other hand, we can bound the same expression by

$$\begin{aligned} |\operatorname{Im} z| |G_z^{(\alpha)}(x, y)|^2 &\leq |\operatorname{Im} z| \sum_{y' \in \mathbb{Z}^d} |G_z^{(\alpha)}(x, y')|^2 \\ &= |\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle \end{aligned}$$

For H_ω bounded and self adjoint, also $H^{(\alpha)}$ is bounded and self adjoint, da $\alpha \in \mathbb{R}$ and $\alpha < \infty$. It holds then,

$$|\operatorname{Im} z| \langle e_x (H^{(\alpha)} - z)^{-1}, (H^{(\alpha)} - z)^{-1} e_x \rangle = |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle$$

Applying the resolvent identity:

$$(H^{(\alpha)} - \bar{z})^{-1} - (H^{(\alpha)} - z)^{-1} = (H^{(\alpha)} - \bar{z})^{-1}(\operatorname{Im}(z))(H^{(\alpha)} - z)^{-1}$$

. We obtain:

$$\begin{aligned} & |\operatorname{Im} z| \langle e_x, (H^{(\alpha)} - \bar{z})^{-1} (H^{(\alpha)} - z)^{-1} e_x \rangle \\ &= |\operatorname{Im} z| \langle e_x, \frac{1}{\operatorname{Im}(z)} \left[(H^{(\alpha)} - z)^{-1} - (H^{(\alpha)} - \bar{z})^{-1} \right] e_x \rangle \\ &= |\operatorname{Im} G^{(\alpha)}(x, x; z)| \\ &= \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2}, \end{aligned} \tag{4.2.14}$$

where the last step used (4.2.12) with $x = y$.

For $t \geq 0$ one has $\min(1, t^2) \leq t^s$, with $s \in (0, 1)$. Using this to interpolate between (4.2.13) and (4.2.14) we get

$$|\operatorname{Im} z| |G^{(\alpha)}(x, y; z)|^2 \leq \frac{|\operatorname{Im} G_\omega(x, x; z)^{-1}|}{|\alpha + G_\omega(x, x; z)^{-1}|^2} \cdot \frac{|G_\omega(x, y; z)|^s}{|G_\omega(x, x; z)|^s}. \tag{4.2.15}$$

We will now use the following “re-sampling trick”, which has the effect of creating an additional random variable (here α) to average over. For a non-negative Borel function f on $\mathbb{R}$,

$$\begin{aligned} \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) d\alpha \rho(\omega_x) d\omega_x &= \iint f(\omega_x + \alpha) \rho(\omega_x + \alpha) \rho(\omega_x) d\omega_x d\alpha \\ &= \iint f(\omega_x) \rho(\omega_x) \rho(\omega_x - \alpha) d\omega_x d\alpha \\ &= \int f(\omega_x) \rho(\omega_x) \left(\int \rho(\omega_x - \alpha) d\alpha \right) d\omega_x \\ &= \int f(\omega_x) \rho(\omega_x) d\omega_x, \end{aligned} \tag{4.2.16}$$

. where in the first and third steps we used Fubini, in the second we used translation invariance of Lebesgue measure and in the fourth we used the fact that $\int \rho(\omega_x - \alpha) d\alpha = 1$.

Choose $f(\omega_x) = |G_z(x, y)|^2$; then (4.2.15) and (4.2.16) yield

$$\begin{aligned} |\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) &= |\operatorname{Im} z| \mathbb{E}_x \left(\int |G_z^{(\alpha)}(x, y)|^2 \rho(\omega_x + \alpha) d\alpha \right) \\ &\leq \mathbb{E}_x \left(|\operatorname{Im} G_z(x, x)|^{-1} \frac{|G_z(x, y)|^s}{|G_z(x, x)|^s} \int \frac{\rho(\omega_x + \alpha)}{|\alpha + G_z(x, x)^{-1}|^2} d\alpha \right). \end{aligned} \quad (4.2.17)$$

We now use Lemma 4.2.8 below with $w = G_z(x, x)^{-1}$ to conclude

$$|\operatorname{Im} z| \mathbb{E}_x(|G_z(x, y)|^2) \leq C_2 \mathbb{E}_x(|G_z(x, y)|^s),$$

with a constant $C_2 < \infty$ which only depends on $\operatorname{supp} \rho$, but not on x, y and z . $\square$

Lemma 4.2.8. *Let $0 < s < 1$. There exists a constant $C_2 := C_2(s) < \infty$ such that*

$$|\operatorname{Im} w| \cdot |w|^s \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \leq C_2$$

uniformly in $w \in \mathbb{C}$ and $\omega_x \sim \mathcal{U}[-1, 1]$.

Proof. The proof is that of [REFERENCE GUNTER STOLZ]. Since $|w|^s \leq |\alpha|^s + |\alpha + w|^s$, we bound:

1.

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq |\operatorname{Im} w| \int \frac{\|\alpha\|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha \\ &\leq |\operatorname{Im} w| \|\alpha\|^s \rho(\omega_x + \alpha) \int \frac{1}{(\alpha + \operatorname{Re} w)^2 + \operatorname{Im} w^2} d\alpha \end{aligned}$$

for $a, b \in \mathbb{R}$ we have

$$\int \frac{1}{a^2 + b^2} da = \frac{1}{b} \left[\arctan \left(\frac{a}{b} \right) \right]_{-\infty}^{\infty} = \frac{\pi}{b}.$$

Thus for $\rho = \frac{1}{2} \chi_{[-1, 1]}$ and $|\alpha| \leq 1 + |\omega| \leq 2$

$$\begin{aligned} |\operatorname{Im} w| \int \frac{|\alpha|^s \rho(\omega_x + \alpha)}{|\alpha + w|^2} d\alpha &\leq \pi \|\alpha\|^s \rho(\omega_x + \alpha) \\ &\leq \pi 2^{s-1}. \end{aligned}$$

2. Now, we bound the second expression.

(a) For $|\operatorname{Im} w|$ big:

Since $|\alpha + w| \geq |\operatorname{Im} w|$ and $\int \rho(\omega_x + \alpha) d\alpha = 1$:

$$I_2 \leq |\operatorname{Im} w| \cdot \frac{1}{|\operatorname{Im} w|^{2-s}} \int \rho(\omega_x + \alpha) d\alpha = \frac{1}{|\operatorname{Im} w|^{1-s}}$$

(b) For $|\operatorname{Im} w|$ small:

$$\begin{aligned} |\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha &\leq |\operatorname{Im} w| \frac{1}{2} \int \frac{d\alpha}{|\alpha + w|^{2-s}} \\ &= |\operatorname{Im} w| \frac{1}{2} \int \frac{1}{|u + \operatorname{Im} w|^{2-s}} du \\ &= |\operatorname{Im} w|^2 \frac{1}{2} \int \frac{1}{|u^2 + \operatorname{Im} w^2|^{\frac{2-s}{2}}} du \\ &= \frac{1}{2} (\operatorname{Im} w)^s \int \frac{1}{|t^2 + 1|^{\frac{2-s}{2}}} dt \end{aligned}$$

by setting $u = \alpha + \operatorname{Re} w$ and $t = \frac{u}{\operatorname{Im} w}$.

To establish the finiteness of the constant, we consider the integral

$$C_s := \int_{-\infty}^{\infty} \frac{1}{(t^2 + 1)^{\frac{2-s}{2}}} dt.$$

For $s \in (0, 1)$, the exponent satisfies $2 - s > 1$, which ensures the asymptotic decay $(t^2 + 1)^{-\frac{2-s}{2}} \sim |t|^{-(2-s)}$ as $|t| \rightarrow \infty$, rendering the integral absolutely convergent. Then we have

$$|\operatorname{Im} w| \int \frac{\rho(\omega_x + \alpha)}{|\alpha + w|^{2-s}} d\alpha \leq \min \left(\frac{1}{|\operatorname{Im} w|^{1-s}}, C_s \frac{1}{2} |\operatorname{Im} w|^s \right) \leq C'_s 2^{s-1}.$$

□

Now we can proceed with the proof of the theorem. The proof, recalls the one proposed in [Sto11]

Proof of Theorem 4.2.5. For H_ω bounded and self-adjoint, from the spectral theorem (see Definition 1.2.1), it exists a complex Borel spectral measure $\mu_{\delta_x, \delta_y} = \mu_{x,y}$ such that:

$$\mu_{x,y}(B) := \langle \delta_x, \chi_B(H_\omega) \delta_y \rangle.$$

with B Borel in $\mathbb{R}$. The *total variation measure* $|\mu_{x,y}|$ is a regular bounded

Borel measure, and satisfies

$$|\mu_{x,y}|(B) = \sup \sum_{i=1}^{\infty} |\mu_{x,y}|(B_j) \quad (4.2.18)$$

where $B = \bigcup B_j$, with B_j disjoint. For $\mu_{x,y}$ complex, there exists α_j complex, such that $|\mu_{x,y}(B_j)| = \alpha_j \mu_{x,y}(B_j)$ and $|\alpha_j| = 1$. Let $g := \sum_{j=1}^{\infty} \alpha_j \chi_{B_j}$, which implies that $|g| \leq 1$. Then it holds:

$$\left| \int_B g(\xi) d\mu_{x,y}(\xi) \right| = \left| \sum_{j=1}^{\infty} \alpha_j \mu_{x,y}(B_j) \right| = \sum_{j=1}^{\infty} |\mu_{x,y}(B_j)|. \quad (4.2.19)$$

From 4.2.18 we obtain:

$$\begin{aligned} |\mu_{x,y}|(B) &= \sup_{\substack{g: \mathbb{R} \rightarrow \mathbb{C} \text{ Borel} \\ |g| \leq 1}} \left| \int_B g(\xi) d\mu_{x,y}(\xi) \right| \\ &= \sup_{|g| \leq 1} |\langle \delta_x, g(H_\omega) \chi_B(H_\omega) \delta_y \rangle|, \end{aligned} \quad (4.2.20)$$

e.g. [Rud87, Chapter 6]. This implies for $g = e^{-itH_\omega}$:

$$\sup_{t \in \mathbb{R}} |\langle \delta_x, e^{-itH_\omega} \chi_B(H_\omega) \delta_y \rangle| \leq |\mu_{x,y}|(B).$$

Therefore, Theorem 4.2.5 will follow from a corresponding exponential decay bound for $\mathbb{E}(|\mu_{x,y}|(B))$.

From Lusin's theorem, we can extend the bound also for $g \in C_c(B)$:

$$|\mu_{x,y}|(B) = \sup_{\substack{g \in C_c(B) \\ |g| \leq 1}} |\langle \delta_x, g(H_\omega) \delta_y \rangle|.$$

for B bounded open interval. For $g \in C_c(B)$ it follows (using that g is bounded and uniformly continuous) that, uniformly in $\xi \in \mathbb{R}$,

$$g(\xi) = \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int \frac{g(E)}{(\xi - E)^2 + \eta^2} dE.$$

Now, always assuming that $g \in C_c(B)$, by the spectral theorem (see Def.

1.2.1), dominated convergence and Fubini:

$$\begin{aligned}
\langle \delta_x, g(H_\omega) \delta_y \rangle &= \int_{\mathbb{R}} g(\xi) d\mu_{x,y}(\xi) \\
&= \lim_{\eta \downarrow 0} \int_{\mathbb{R}} \left(\frac{\eta}{\pi} \int_B \frac{g(E)}{(\xi - E)^2 + \eta^2} dE \right) d\mu_{x,y}(\xi) \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \left(\int_{\mathbb{R}} \frac{1}{(\xi - E)^2 + \eta^2} d\mu_{x,y}(\xi) \right) dE \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, ((H_\omega - E)^2 + \eta^2)^{-1} \delta_y \rangle dE \\
&= \lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE
\end{aligned}$$

Now we can bound $\mathbb{E}(|\mu_{x,y}|)$, considering that $|g(E)| \leq 1$:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(B)) &\leq \mathbb{E} \left(\lim_{\eta \downarrow 0} \frac{\eta}{\pi} \int_B g(E) \langle \delta_x, (H_\omega - E - i\eta)^{-1} (H_\omega - E + i\eta)^{-1} \delta_y \rangle dE \right) \\
&\leq \mathbb{E} \left(\lim_{\eta \rightarrow 0^+} \frac{\eta}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle| |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle| dE \right) \\
&\leq \lim_{\eta \rightarrow 0^+} \frac{1}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} (\mathbb{E}(\eta |\langle \delta_x, (H_\omega - E - i\eta)^{-1} \delta_z \rangle|^2))^{1/2} \\
&\quad \cdot (\mathbb{E}(\eta |\langle \delta_z, (H_\omega - E + i\eta)^{-1} \delta_y \rangle|^2))^{1/2} dE,
\end{aligned}$$

where, in this order, Fatou, Fubini and Cauchy-Schwarz (on $\mathbb{E}$) have been used. Now we can use Proposition 4.2.7 and the hypothesis to obtain:

$$\begin{aligned}
\mathbb{E}(|\mu_{x,y}|(B)) &\leq \frac{C_1}{\pi} \int_B \sum_{z \in \mathbb{Z}^d} (\mathbb{E}|G_\omega(x, z; E + i\eta)|^s)^{1/2} (\mathbb{E}|G_\omega(z, y; E - i\eta)|^s)^{1/2} dE \\
&\leq \frac{C_1|B|C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{2}|x-z|} e^{-\frac{\mu}{2}|z-y|}.
\end{aligned}$$

By the triangle inequality, $\frac{\mu}{2}|x-z| + \frac{\mu}{2}|z-y| \geq \frac{\mu}{4}|x-y| + \frac{\mu}{4}|x-z| + \frac{\mu}{4}|z-y|$. Therefore,

$$\mathbb{E}(|\mu_{x,y}|(B)) \leq \left(\frac{C_1|B|C}{\pi} \sum_{z \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|x-z|} e^{-\frac{\mu}{4}|z-y|} \right) e^{-\frac{\mu}{4}|x-y|} \leq \tilde{C} e^{-\tilde{\mu}|x-y|},$$

with $\tilde{\mu} = \frac{\mu}{4}$ and finite prefactor $\tilde{C} = \frac{C_1|B|C}{\pi} \sum_{u \in \mathbb{Z}^d} e^{-\frac{\mu}{4}|u|}$. $\square$

From (3.2.2) we can also directly obtain spectral localization through Simon-Wolff criterion.

Proposition 4.2.9 (Simon-Wolff criterion). *Let H_ω be a random operator in $l^2(\mathbb{Z}^d)$ with a random potential whose conditional-single site distribution is absolutely continuous. Then if for Lebesgue-almost every $E \in \mathbb{R}$ and $\mathbb{P}$ -almost all ω :*

$$\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 < \infty \quad (4.2.21)$$

H_ω has only pure point spectrum.

Proof can be found in [AW15b, Chapter 5]

Theorem 4.2.10. *Let Theorem 4.2.1 hold. Then, the spectrum of H_ω consists only of pure point spectrum for a.e. realisation of V_ω .*

Proof from [DERM24]. From (4.2.1) we notice that for all $\lambda > \lambda_{\text{And}}$:

$$\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] = C \sum_{y \in \mathbb{Z}^d} e^{-\mu|x-y|} < \infty \quad \text{for all } \mu > 0 \quad (4.2.22)$$

uniformly for $z \in \mathbb{C} \setminus \mathbb{R}$ and $x \in \mathbb{Z}^d$ fixed.

From Proposition 4.2.7 we notice also that $\sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^2]$ always exists but might be infinite for $\eta \rightarrow 0$. We argue:

$$\begin{aligned} \mathbb{E} \left[\left(\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] &= \mathbb{E} \left[\lim_{\eta \downarrow 0} \left(\sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^2 \right)^{\frac{s}{2}} \right] \\ &\leq \mathbb{E} \left[\lim_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} |G_{E+i\eta}(x, y)|^s \right] \\ &\leq \liminf_{\eta \downarrow 0} \sum_{y \in \mathbb{Z}^d} \mathbb{E}[|G_{E+i\eta}(x, y)|^s] < \infty, \end{aligned}$$

where in the first two steps we used that the function $x \mapsto x^{\frac{s}{2}}$ is monotone and $|\sum_n a_n|^s \leq \sum_n |a_n|^s$ for all $a_n \geq 0$ and $0 < s < 1$. The last two inequalities follow by Fatou and (4.2.22). $\square$

4.2.2 Exponential decay of the eigenfucntions

From *exponential and dynamical localization*, we can obtain that all the eigenfunction of the time-evolving Schrödinger operator decay also exponentially.

1. da simon wolff ottengo Theorem 7.4
2. dalla dynamical ottengo che decade più veloce di ogni polinomio
3. dalla exponential so che la sua time evolution

Definition 5.1.2 (Sets of Self-Avoiding Walks). Let $\Lambda \subseteq \mathbb{Z}^d$. For any $x, y \in \Lambda$ and $N \in \mathbb{N}_0$:

- (i) $\mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N contained entirely in Λ , starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. When $\Lambda = \mathbb{Z}^d$, we simply write $\mathcal{S}_N(y, x)$.
- (ii) $\mathcal{S}_N^{(\Lambda)}(x) := \bigcup_{y \in \Lambda} \mathcal{S}_N^{(\Lambda)}(y, x)$ denotes the set of all self-avoiding walks of length N in Λ starting at x .
- (iii) $c_N := |\mathcal{S}_N(x)| = \sum_{y \in \mathbb{Z}^d} |\mathcal{S}_N(y, x)|$ denotes the total number of self-avoiding walks of length N on $\mathbb{Z}^d$ originating from a fixed site (e.g., the origin 0), which is independent of the starting vertex by translation invariance.

Definition 5.1.3 (Connectivity Constant). The **connective constant** (or connectivity constant) μ_d of the lattice $\mathbb{Z}^d$ is defined as the asymptotic growth rate of c_N :

$$\mu_d := \lim_{N \rightarrow \infty} (c_N)^{1/N}.$$

By subadditivity of $\log c_N$, this limit exists, is positive, and satisfies the strict inequality $\mu_d < 2d - 1$.

Claim 5.1.4. Let μ_d be the connective constant (or growth rate) of self-avoiding walks on $\mathbb{Z}^d$. Then:

$$\mu_d \leq 2d - 1$$

Proof. Let c_N denote the number of self-avoiding walks (SAWs) of length N starting from the origin $0 \in \mathbb{Z}^d$, and let $S_N(0, x)$ denote the set of such walks ending at $x \in \mathbb{Z}^d$. By definition:

$$c_N = \sum_{x \in \mathbb{Z}^d} |S_N(0, x)| = \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in S_N(0, x)} 1$$

Consider the generating function (*susceptibility*) with parameter $\xi \in \mathbb{C}$:

$$\sum_{N=0}^{\infty} |\xi|^N c_N = \sum_{N=0}^{\infty} |\xi|^N \sum_{x \in \mathbb{Z}^d} |S_N(0, x)|$$

For $N = 0$, $c_0 = 0$. For $N \geq 1$, a walk cannot immediately step back onto the previous vertex (non-reversing condition). Thus:

- The first step has at most $2d$ choices.
- Each of the subsequent $N - 1$ steps has at most $2d - 1$ choices.

This gives the upper bound:

$$c_N \leq 2d(2d - 1)^{N-1}, \quad \text{for } N \geq 1.$$

Using this upper bound, we can bound the series from above:

$$\begin{aligned} \sum_{N=0}^{\infty} |\xi|^N c_N &\leq \sum_{N=1}^{\infty} |\xi|^N (2d)(2d - 1)^{N-1} \\ &= 2d|\xi| \sum_{k=0}^{\infty} |\xi|^k (2d - 1)^k \quad (\text{setting } k = N - 1) \\ &= 2d|\xi| \sum_{k=0}^{\infty} (|\xi|(2d - 1))^k \end{aligned}$$

This geometric series converges if and only if:

$$|\xi|(2d - 1) < 1 \iff |\xi| < \frac{1}{2d - 1} =: R$$

where $R = \frac{1}{2d-1}$ is the radius of convergence of the bounding series.

By the Cauchy–Hadamard theorem, the radius of convergence R' of the original series $\sum_{N=0}^{\infty} c_N |\xi|^N$ is:

$$R' = \frac{1}{\limsup_{N \rightarrow \infty} c_N^{1/N}} = \frac{1}{\mu_d}$$

Since the original series is bounded term-by-term by a series that converges for $|\xi| < R$, the true radius of convergence must satisfy:

$$R' \geq R \implies \frac{1}{\mu_d} \geq \frac{1}{2d - 1}$$

Taking the reciprocal yields:

$$\mu_d \leq 2d - 1.$$

□

Definition 5.1.5 (SAW Correlation Function). For a parameter $\beta \geq 0$, the **self-avoiding walk correlation function** $C_\beta(y - x)$ between

two vertices $x, y \in \mathbb{Z}^d$ is the generating function

$$C_\beta(y - x) := \sum_{N=0}^{\infty} \beta^N |\mathcal{S}_N(y, x)|,$$

defined for all values of β for which the power series is absolutely convergent. By translation invariance, $C_\beta(y - x)$ depends only on the displacement vector $y - x$.

Definition 5.1.6 (SAW Susceptibility). The **susceptibility** $\chi(\beta)$ of self-avoiding walks is the sum of the two-point correlation function over all end-points:

$$\chi(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta(x) = \sum_{N=0}^{\infty} c_N \beta^N.$$

The series converges uniformly and absolutely if and only if $(c_N)^{1/N} \beta < 1$, where the critical radius of convergence R_χ is given by $\lim_{N \rightarrow \infty} \frac{1}{(c_N)^{1/N}} = \frac{1}{\mu_d}$.

Proposition 5.1.7 (Exponential Decay of the Correlation Function). *Let $\beta \geq 0$ be such that $\beta < 1/\mu_d$. Then for every $\varepsilon > 0$ there exists a constant $K_\varepsilon < \infty$, independent of x , such that*

$$C_\beta(x) \leq K_\varepsilon ((\mu_d + \varepsilon)\beta)^{|x|} \quad \text{for all } x \in \mathbb{Z}^d.$$

Proof. We know that $|\mathcal{S}_N(0, x)| = 0$ whenever $N < |x|$, and the sum defining $C_\beta(x)$ may be restricted to $N \geq |x|$:

$$C_\beta(x) = \sum_{N=|x|}^{\infty} \beta^N |\mathcal{S}_N(x, 0)| \leq \sum_{N=|x|}^{\infty} \beta^N c_N. \quad (5.1.1)$$

Fix $\varepsilon > 0$. Since $\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$, by the definition of limit there exists $N_0 \in \mathbb{N}$ such that

$$c_N^{1/N} < \mu_d + \varepsilon \quad \text{for all } N \geq N_0,$$

that is,

$$c_N < (\mu_d + \varepsilon)^N \quad \text{for all } N \geq N_0.$$

For $N = 0, 1, \dots, N_0 - 1$, define

$$K_\varepsilon := \max \left(1, \max_{0 \leq N < N_0} \frac{c_N}{(\mu_d + \varepsilon)^N} \right),$$

which is finite, being the maximum of finitely many finite quantities. By construction it holds:

$$c_N \leq K_\varepsilon (\mu_d + \varepsilon)^N \quad \text{for all } N \geq 0.$$

Combining it with (4.1) we obtain

$$C_\beta(x) \leq K_\varepsilon \sum_{N=|x|}^{\infty} [(\mu_d + \varepsilon)\beta]^N.$$

Choosing ε small enough that $r := (\mu_d + \varepsilon)\beta < 1$ (possible since $\beta < 1/\mu_d$), the tail of the geometric series equals

$$\sum_{N=|x|}^{\infty} r^N = \frac{r^{|x|}}{1 - r}.$$

Hence

$$C_\beta(x) \leq \frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta} [(\mu_d + \varepsilon)\beta]^{|x|}.$$

where $\frac{K_\varepsilon}{1 - (\mu_d + \varepsilon)\beta}$ is still finite and independent from x . $\square$

5.2 SAW representation of G

The proof of the theorem 3.1 is similar to the one given by Schenker in [1]. We are going to work with the Green function defined on a volume where one site is removed every stage. Therefore, for any $\Lambda \subseteq \mathbb{Z}^d$ (finite or infinite) the Green function changes:

$$G_z^\Lambda(x, y) := \begin{cases} (H_\omega^\Lambda - z), & \forall x, y \in \Lambda \\ 0, & \text{otherwise} \end{cases} \quad (5.2.1)$$

where $H_\omega^\Lambda - z \in \mathbb{C}^{\Lambda \times \Lambda}$ is the matrix obtained by restricting $H_\omega - z$ to Λ . In this thesis we will consider $\Lambda = \mathbb{Z}^d$, but we leave the notation Λ to highlight the fact that the same result holds for any volume.

To notice is also that $H_\omega^\Lambda - z$ is still invertible for all $z \in \mathbb{C} \setminus \mathbb{R}$. In this section, for simplicity during calculations, we will write $G^\Lambda(x, y)$ instead of $G_z^\Lambda(x, y)$.

Theorem 5.2.1. *Let $x, y \in \mathbb{Z}^d$, $x \neq y$, $\Lambda \subseteq \mathbb{Z}^d$. G^Λ is defined as in (4.1).*

It holds

$$G^\Lambda(x, y) = \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{i=1}^n G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) + R_N \quad (5.2.2)$$

where $\Lambda_i = \Lambda \setminus \{x, \gamma_1, \dots, \gamma_{i-1}\}$ and

$$R_N = \sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \prod_{i=1}^{N-1} G^\Lambda(x, x) \cdot G^{\Lambda_i}(\gamma_i, \gamma_i) \cdot G^{\Lambda_N}(\gamma_N, y). \quad (5.2.3)$$

In order to prove Theorem 4.8,

Claim 5.2.2.

$$G^\Lambda(x, y) = G^\Lambda(x, x) \sum_{x_1 \sim x} G^{\Lambda_1}(x_1, y) \quad (5.2.4)$$

Proof. Using the resolvent identity for $A = (H_\omega^\Lambda - z)$ and $B = (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)$, we obtain:

$$\begin{aligned} & (H_\omega^\Lambda - z)^{-1} - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \\ &= -(H_\omega^\Lambda - z)^{-1} \left(H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) \right) (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \end{aligned}$$

Moreover:

$$H_\omega^\Lambda - (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}}) = - \sum_{x_1 \sim x} \delta_{x_1} =: T_{x, x_1}$$

Now considering the matrix elements in (x, y) with $x \neq y$ and matrix multiplication:

$$\langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_y \rangle - 0 = - \sum_{x_1, x_2} \langle \delta_x, (H_\omega^\Lambda - z)^{-1} \delta_{x_1} \rangle T_{x_1, x_2} \langle \delta_{x_2}, (H_\omega^{\{x\}} \oplus H_\omega^{\Lambda \setminus \{x\}} - z)^{-1} \delta_y \rangle$$

For $T_{x_1, x_2} \neq 0$, we need $x_1 = x$ and $x_2 \sim x_1$.

Which leaves us with:

$$G^\Lambda(x, y) = \sum_{x_1 \sim x} G^\Lambda(x, x) G^{\Lambda \setminus \{x\}}(x_1, y)$$

□

Proof of Theorem 5.8. Now we separate the case in which $x_1 = y$ (which

implies $|x - y| = 1$).

$$G^\Lambda(x, y) = G^\Lambda(x, x) \left(\sum_{x_1 \sim x, x_1 \neq y} G^{\Lambda_1}(x_1, y) + \delta_{|x-y|=1} G^{\Lambda_1}(y, y) \right)$$

Iterating N times, it gives (5.2.2). $\square$

5.3 Proof with SAW

In this last Section, we can use the SAW representation of the Green function (5.2.2), in order to prove Theorem 4.2.1.

Proof. We take the average of (4.2), and using $|\sum a|^s \leq \sum |a|^s$ (which follows from triangle inequality), we have:

$$\mathbb{E} [|G^\Lambda(x, y)|^s] \leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x, y)} \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{i=1}^n |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \right] + \quad (5.3.1)$$

$$\sum_{x \sim y} \sum_{\gamma \in \mathcal{S}_N(0, x)} \mathbb{E} [|G^\Lambda(x, x)|^s \prod_{i=1}^{N-1} |G^{\Lambda_i}(\gamma_i, \gamma_i)|^s \cdot |G^{\Lambda_N}(\gamma_N, y)|^s] \quad (5.3.2)$$

The resolvent $G^{\Lambda_j}(w, w')$ does not depend on the random variables ω_{x_i} , $i = 0, \dots, j-1$. Hence, recursive applications of the a priori bound (3.2) yield

$$\mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \right] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^{n+1}$$

and

$$\begin{aligned} & \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \cdot |G^{\Lambda_{N-1}}(\gamma_N, y)|^s \right] \\ & \leq \left(\frac{1}{1-s} \cdot \frac{1}{\lambda^s} \right)^N \cdot \mathbb{E} [|G^{\Lambda_{N-1}}(\gamma_N, y)|^s] \leq \left(\frac{1}{1-s} \frac{1}{\lambda^s} \right)^N \frac{1}{|\operatorname{Im}(z)|^s} \end{aligned}$$

Where we used a priori bound and lemma 3.5.

Inserting these estimates in the sums above, we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^N \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| + \text{Err}(N) \frac{1}{|\text{Im} z|^s} \quad (5.3.3)$$

with $\text{Err}(N) := \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)|$ $\beta(s) = \frac{1}{1-s} \frac{1}{\lambda^s}$ Let's consider the susceptibility $\chi(\beta)$ of the self avoiding walks in $\mathcal{S}_N^{(\Lambda)}(x)$:

$$\chi(\beta) = \sum_{x \in \mathbb{Z}^d} \sum_{N=0}^{\infty} \beta(s)^N |\mathcal{S}_N^{(\Lambda)}(x)| = \sum_{N=0}^{\infty} c_N \beta(s)^N. \quad (5.3.4)$$

The series converges absolutely if and only if $0 \leq |\beta| < 1/\mu_d$, where $1/\mu_d = \lim_{N \rightarrow \infty} c_N^{1/N}$ is the critical radius of convergence. Thus, for $|\beta(s)| = \frac{1}{|(1-s)\lambda^s|} \leq \frac{1}{\mu_d}$, $\chi(\beta)$ converges uniformly, and most importantly:

$$\text{Err}(N) \longrightarrow 0, \quad \text{for } N \rightarrow \infty$$

We send N to the infinity and obtain from (4.7):

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \sum_{n=0}^{\infty} \beta(s)^{1+n} |\mathcal{S}_n^{(\Lambda)}(y, x)| \quad (5.3.5)$$

Taking the supremum over volumes Λ now yields

$$\sup_{\Lambda \subset \mathbb{Z}^d} \mathbb{E} \left(|G^{(\Lambda)}(x, y)|^s \right) \leq \beta(s) C_{\beta(s)}(x - y). \quad (4.3)$$

It remains to establish for which λ we may find $s \in (0, 1)$ with $\beta(s) < 1/\mu_d$. As above, for $\lambda > e$ the unique critical point is $s_{\text{crit}} = 1 - 1/\ln \lambda$ and

$$\min_{s \in (0, 1)} \beta(s) = \beta(s_{\text{crit}}) = \frac{e \ln \lambda}{\lambda}$$

which is less than $1/\mu_d$ if and only if $\lambda > \lambda_{\text{And}}$. Finally we get:

$$\mathbb{E} \left(|G^{(\Lambda)}(x, y; z)|^{1 - \frac{1}{\ln \lambda}} \right) \leq \beta(\lambda) K_{\epsilon} \ln \lambda e^{-m_{\epsilon}(\lambda)|x-y|} \quad (5.3.6)$$

where

$$m_{\epsilon}(\lambda) = -\ln \beta(\lambda) - \ln(\mu_d + \epsilon) > 0.$$

□

Chapter 6

Connection to Long-Range SAW

The purpose of this chapter is to extend the self-avoiding walk representation of the resolvent, to random Hamiltonians with long-range hopping. In particular, we compare this setting with the previous model and apply the fractional moment method to study localization.

The difference lies in the kinetic operator, which is replaced by the fractional discrete Laplacian $(-\Delta)^\alpha$ with $0 < \alpha < 1$. This change leads to a polynomial decay bound for the averaged Green function, in sharp contrast with the standard exponential decay.

This chapter is based on the work of M. Disertori, R.M. Escobar and C. Rojas-Molina in 2024 [DERM24].

6.1 The fractional Laplacian

The fractional kinetic operator also connects arbitrarily distant sites in the lattice, rather than just the first neighbors.

Definition 6.1.1. Let $-\Delta$ be defined as in Section 1.1. The fractional discrete Laplacian is $(-\Delta)^\alpha$ for $0 < \alpha < 1$

Just as the standard discrete Laplacian, $(-\Delta)^\alpha$ is bounded, self-adjoint and translation invariant. Its entries satisfy

$$(-\Delta)^\alpha(x, x) > 0 \quad \forall x \in \mathbb{Z}^d, \quad (-\Delta)^\alpha(x, y) < 0 \quad \forall x \neq y \text{ in } \mathbb{Z}^d$$

Moreover, its off-diagonal matrix elements exhibit polynomial decay: there exist constants $0 < c_{\alpha,d} \leq C_{\alpha,d} < \infty$ such that

$$\frac{c_{\alpha,d}}{|x-y|^{d+2\alpha}} \leq -(-\Delta)^\alpha(x,y) \leq \frac{C_{\alpha,d}}{|x-y|^{d+2\alpha}} \quad (6.1.1)$$

We notice that for $x \neq y$, the fractional Laplacian is never exactly equal to zero. This characteristic describes the long-jump behavior: now, the discrete Laplacian reaches every site of the grid, since the values are never zero, in contrast with the standard case, where the only value different from zero were on the diagonal and first neighbor sites.

Moreover, (6.1.1) confirms the boundness of the fractional operator for all $x \neq y$ and the summability of its kernel $\sum_{x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)| < \infty$. For the diagonal entries, it holds:

$$(-\Delta)^\alpha(x,x) = - \sum_{y \neq x} (-\Delta)^\alpha(x,y)$$

which confirms boundness also for $x = y$. Notice that for $\alpha = 1$, we can use the same sum and obtain $(-\Delta)(x,x) = - \sum_y (-\Delta)(x,y) = 2d$.

We can already prove the summability of the fractional kernel too:

Lemma 6.1.2. *For any $\alpha \in (0,1)$ and $s \in \left(\frac{d}{d+2\alpha}, 1\right]$,*

$$\sum_{x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)|^s < \infty \quad (6.1.2)$$

Proof. From (6.1.1):

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} |(-\Delta)^\alpha(0,x)|^s \leq C_{\alpha,d}^s \sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^{(d+2\alpha)s}}.$$

By standard comparison of the lattice sum with an integral on $\mathbb{R}^d$:

$$\sum_{|x| \geq 1, x \in \mathbb{Z}^d} \frac{1}{|x|^p} \leq C_d \int_{|z| > 1} \frac{1}{|z|^p} dz.$$

Passing to spherical coordinates in $\mathbb{R}^d$, the integral reads

$$\begin{aligned} \int_{|z|>1} \frac{1}{|z|^p} dz &= |\mathbb{S}^{d-1}| \int_1^\infty \frac{r^{d-1}}{r^p} dr \\ &= |\mathbb{S}^{d-1}| \int_1^\infty r^{d-1-p} dr, \end{aligned}$$

where $|\mathbb{S}^{d-1}| = \frac{2\pi^{d/2}}{\Gamma(d/2)}$ denotes the surface area of the unit sphere in $\mathbb{R}^d$.

The radial integral converges if and only if the power satisfies

$$d - 1 - p < -1 \iff p > d.$$

Substituting $p = s(d + 2\alpha)$, this condition becomes

$$s(d + 2\alpha) > d \iff s > \frac{d}{d + 2\alpha}.$$

Since $\alpha > 0$ implies $d + 2\alpha > d$; for every $s \in \left(\frac{d}{d+2\alpha}, 1\right]$, we conclude that

$$\sum_{x \neq 0} |(-\Delta)_{0,x}^\alpha|^s \leq C_{\alpha,d}^s \sum_{x \neq 0} \frac{1}{|x|^{s(d+2\alpha)}} < \infty.$$

□

6.2 The SAW Representation for Long-Range Resolvents

We consider the fractional Hamiltonian

$$H_{\alpha,\omega} = (-\Delta)^\alpha + \lambda V_\omega \quad \text{on } \ell^2(\mathbb{Z}^d), \quad (6.2.1)$$

where $0 < \alpha < 1$ and V_ω is a diagonal matrix with i.i.d. random entries defined as in Section 1.1. The Green function $G_z = (H_{\alpha,\omega} - z)^{-1}$ is well-defined for all $z \in \mathbb{C} \setminus \mathbb{R}$, and its decay at strong disorder is closely related to the polynomial decay of the fractional Laplacian.

As before, in the long-range scenario, we use the geometric resolvent expansion to represent the fractional moments as a sum over self-avoiding

paths and we obtain that the averaged fractional Green function satisfies

$$\mathbb{E}[|G_z(x, y)|^s] \leq \frac{K_0}{(1-s)\lambda^s} \frac{1}{|x-y|^{d+2\alpha_s}} \quad \forall x, y \in \mathbb{Z}^d, x \neq y, \quad (6.2.2)$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$, for all $\lambda > \lambda_0(s) := \left(\frac{1}{(1-s)R_{\chi_D}} \right)^{\frac{1}{s}}$. Here, $2\alpha_s := s(d+2\alpha) - d$ and $\frac{1}{(1-s)\lambda^s}$ comes from the *a priori bound*.

Since we are considering arbitrary sites on the lattice, we define the step weight kernel for the SAW as $D(x, y) := |(-\Delta)^\alpha(x, y)|^s$, which generates a transition probability $p(x \rightarrow y) := \frac{D(x, y)}{\sum_{z \neq x} D(x, z)}$ and is translation-invariant. This is well-defined for $s \in \left(\frac{d}{d+2\alpha}, 1 \right)$ because the kernel is summable: $\sum_{x \neq 0} |(-\Delta)^\alpha(0, x)|^s < \infty$ (See (6.1.2)). From this essential condition, we develop the same strategy as for the standard Anderson resolvent. The new definitions for SAWs are all generated by D .

SAW definitions for the discrete Anderson model. Recall the definitions in Chapter 4. A self-avoiding walk (SAW) on the lattice $\mathbb{Z}^d$ with long-range jumps is characterized by the following essential components:

- $\mathcal{S}_N(x, y)$ denotes the set of all self-avoiding walks of length N starting at $\gamma_0 = x$ and terminating at $\gamma_N = y$. Due to the translation invariance of the fractional laplacian we know $\mathcal{S}_N(x, y) = \mathcal{S}_N(0, y-x)$ for all $x, y \in \mathbb{Z}^d$.
- **Two-point Function:**

$$C_\beta^D(x, y) = C_\beta^D(y-x) := \sum_{n \geq 0} \beta^n \sum_{\gamma \in \mathcal{S}_n(x, y)} \prod_{j=0}^{n-1} D(\gamma_j, \gamma_{j+1}),$$

for all $x, y \in \mathbb{Z}^d$.

- **Susceptibility:**

$$\chi^D(\beta) := \sum_{x \in \mathbb{Z}^d} C_\beta^D(x),$$

which is finite for values strictly below its radius of convergence R_{χ^D} .

In particular, we also consider the following consequences:

Lemma 6.2.1. *Consider the definitions given above. Then it holds*

$$R_\chi^D \geq \frac{1}{\sum_{w \neq 0} D(0, w)} > 0.$$

Proof. For $n = 2$, we have

$$\begin{aligned} \sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_2(0, x)} D(0, \gamma_1) D(\gamma_1, x) &= \sum_{\gamma_1 \neq 0} D(0, \gamma_1) \sum_{x \neq 0, \gamma_1} D(\gamma_1, x) \\ &\leq \sum_{\gamma_1 \neq 0} D(0, \gamma_1) \sum_{x \neq \gamma_1} D(\gamma_1, x) = \left(\sum_{w \neq 0} D(0, w) \right)^2. \end{aligned}$$

Repeating this argument inductively for any general length $n \geq 1$, we obtain

$$\sum_{x \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_n(0, x)} \prod_{j=0}^{n-1} D(\gamma_j, \gamma_{j+1}) \leq \left(\sum_{z \neq 0} D(0, z) \right)^n.$$

By substituting this bound into the definition of the susceptibility, we get a geometric series that converges as long as $\beta \sum_{z \neq 0} D(0, z) < 1$. This guarantees that the radius of convergence satisfies

$$R_{\chi^D} \geq \frac{1}{\sum_{z \neq 0} D(0, z)} > 0. \quad (3.7)$$

□

Proposition 6.2.2. *Let $s \in (\frac{d}{d+2\alpha}, 1)$. Then*

$$C_\beta^D(x, y) \leq K_0 \frac{1}{|y - x|^{d+2\alpha_s}}$$

for all $x, y \in \mathbb{Z}^d$.

Proof in [DERM24, Lemma 4].

As in the previous Chapter, we prove (6.2.2) using a SAW representation of $G_{\alpha, \omega}$.

Theorem 6.2.3. *Let $s \in (\frac{d}{d+2\alpha}, 1)$. Then, for all the above definitions, it holds*

$$\mathbb{E}(|G_{\alpha, \omega}(x, y)|^s) \leq \frac{1}{\lambda^s(1-s)} C_\beta^D(y - x) + \frac{1}{|\text{Im}(z)|^s} R_N \quad (6.2.3)$$

for all $x, y \in \mathbb{Z}^d$. Moreover,

$$R_N := \left(\frac{1}{(1-s)\lambda^s} \right)^N \sum_{\gamma \in \mathcal{S}_{N+1}(x,y)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}). \quad (6.2.4)$$

Proof. As in Chapter 4, we must work with Green functions defined on $\Lambda \subset \mathbb{Z}^d$, and to simplify the notation, we also set $\Delta^\alpha(x, y) := -(-\Delta)^\alpha(x, y)$, which guarantees $\Delta^\alpha(x, y) > 0$ for all $x \neq y$ and $G_{\alpha,z}^\Lambda(x, y) = G^\Lambda(x, y)$. Recall also $\Lambda \setminus \{\gamma_0, \dots, \gamma_{j-1}\} = \Lambda_j$.

By the resolvent identity, we have for all $x \neq y \in \Lambda$

$$\begin{aligned} G^\Lambda(x, y) &= G^\Lambda(x, x) \Delta^\alpha(x, y) G^{\Lambda \setminus \{x\}}(y, y) \\ &\quad + G^\Lambda(x, x) \sum_{\gamma_1 \in \Lambda \setminus \{x, y\}} \Delta^\alpha(x, \gamma_1) G^{\Lambda \setminus \{x\}}(\gamma_1, y). \end{aligned}$$

Repeating this expansion N times and organizing the terms using the sets of self-avoiding walks $\mathcal{S}_n(x, y)$, we obtain

$$\begin{aligned} G^\Lambda(x, y) &= G^\Lambda(x, x) \left[\sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x,y)} \prod_{j=0}^{n-1} \Delta^\alpha(\gamma_j, \gamma_{j+1}) \prod_{j=1}^n G^{\Lambda_1}(\gamma_j, \gamma_j) \right. \\ &\quad \left. + \sum_{\gamma \in \mathcal{S}_{N+1}(x,y)} \prod_{j=0}^{N-1} \Delta^\alpha(\gamma_j, \gamma_{j+1}) \prod_{j=1}^{N-1} G^{\Lambda_j}(\gamma_j, \gamma_j) G^{\Lambda_N}(\gamma_N, y) \right]. \end{aligned}$$

Taking the fractional expectation $\mathbb{E}[\cdot |^s]$ and utilizing the concavity of the function $v \mapsto v^s$ for $0 < s < 1$, we can pass the fractional power inside the sums

$$\begin{aligned} \mathbb{E}[|G^\Lambda(x, y)|^s] &\leq \sum_{n=1}^N \sum_{\gamma \in \mathcal{S}_n(x,y)} \prod_{j=0}^{n-1} \Delta^\alpha(\gamma_j, \gamma_{j+1})^s \cdot \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_1}(\gamma_j, \gamma_j)|^s \right] \\ &\quad + \sum_{\gamma \in \mathcal{S}_{N+1}(x,y)} \prod_{j=0}^{N-1} \Delta^\alpha(\gamma_j, \gamma_{j+1})^s \\ &\quad \cdot \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s |G^{\Lambda_N}(\gamma_N, y)|^s \right]. \end{aligned}$$

The resolvent G^{Λ_j} does not depend on the random variables ω_{γ_i} for $i =$

$0, \dots, j-1$. Hence, we can apply the apriori bound recursively, which yields

$$\begin{aligned} \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^n |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s \right] &\leq \left(\frac{1}{(1-s)\lambda^s} \right)^{n+1}, \\ \mathbb{E} \left[|G^\Lambda(x, x)|^s \prod_{j=1}^{N-1} |G^{\Lambda_j}(\gamma_j, \gamma_j)|^s |G^{\Lambda_{N+1}}(\gamma_N, y)|^s \right] &\leq \left(\frac{1}{(1-s)\lambda^s} \right)^N \frac{1}{|\text{Im}(z)|^s}, \end{aligned}$$

where the deterministic bound $|G^{\Lambda_{N+1}}(\gamma_N, y)| \leq \frac{1}{|\text{Im}(z)|}$ is applied for the remainder term.

Setting the parameter $\beta = \frac{1}{(1-s)\lambda^s}$ and substituting $D(\gamma_j, \gamma_{j+1}) = \Delta^\alpha(\gamma_j, \gamma_{j+1})^s$, we utilize the translation invariance to shift the paths to originate at 0. Inserting the recursive estimates into the summation produces the bound

$$\begin{aligned} \mathbb{E}[|G^\Lambda(x, y)|^s] &\leq \beta \sum_{n=1}^N \beta^n \sum_{\gamma \in \mathcal{S}_n(0, y-x)} \prod_{j=0}^{n-1} D(\gamma_j, \gamma_{j+1}) \\ &\quad + \beta^N \frac{1}{|\text{Im}(z)|^s} \sum_{\gamma \in \mathcal{S}_{N+1}(0, y-x)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}) \\ &\leq \beta C_\beta^D(y-x) + \frac{1}{|\text{Im}(z)|^s} R_N, \end{aligned}$$

□

Finally, the rest term R_N satisfies

$$\begin{aligned} R_N &:= \beta^N \sum_{\gamma \in \mathcal{S}_{N+1}(0, y-x)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}) \\ &\leq \beta^N \sum_{w \in \mathbb{Z}^d} \sum_{\gamma \in \mathcal{S}_N(0, w)} \prod_{j=0}^{N-1} D(\gamma_j, \gamma_{j+1}). \end{aligned}$$

The assumption that $\lambda > \lambda_0(s) := (\frac{1}{(1-s)} \frac{1}{R_{\chi^D}})^{\frac{1}{s}}$ ensures that $\beta < R_{\chi^D}$. Since the susceptibility $\chi^D(\beta)$ is finite within its radius of convergence, taking the limit implies $\lim_{N \rightarrow \infty} R_N = 0$, which, together with (6.2.2), gives us (6.2.2).

6.3 The Localization Criteria

A natural question that may arise is how localization is proven without the exponential decay of the fractional moments of the Green function. Due to the polynomial tail of the fractional Laplacian, exponential decay is not expected; instead, it is only possible to prove a weaker version of dynamical localization where only moments $\beta < 2\alpha$ of the position operator are finite. Spectral localization is proven via the Simon-Wolff theorem.

Lemma 6.3.1. *Let (6.2.2) hold uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for $\lambda > \lambda_0(s)$. It follows that:*

- (i) *The spectrum of $H_{\alpha,\omega}$ consists only of pure point spectrum for a.e. $\omega \in \Omega$.*
- (ii) *For a.e. $\omega \in \Omega$ and for all E in the almost sure spectrum, the corresponding normalized eigenfunction φ_E satisfies:*

$$|\varphi_E(y, \omega)|^2 \leq \frac{C_t(\omega)}{(1 + |y - x_E(\omega)|)^t} \quad \forall 0 < t < 2\alpha_s, \quad (6.3.1)$$

where $x_E(\omega)$ is a localization center and $C_t(\omega)$ is an integrable random variable.

- (iii) *For $0 < \beta < 2\alpha_s$, we have for a.e. $\omega \in \Omega$*

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha,\omega}}(0, x)|^2 < \infty. \quad (6.3.2)$$

In particular, if $\alpha > \frac{1}{2}$, the first moment of the wave function initially localized at the origin and evolving under the action of $H_{\alpha,\omega}$ is finite.

Proof. (i) From (6.2.2), we verify the summability of the Green function moments:

$$\sum_{x \in \mathbb{Z}^d} \mathbb{E}[|G_z(0, x)|^s |x|^t] \leq \frac{K_0 \theta_s}{\lambda^s} \sum_{x \neq 0} \frac{1}{|x|^{d+2\alpha_s-t}} < \infty \quad \forall t \in [0, 2\alpha_s),$$

uniformly in $z \in \mathbb{C} \setminus \mathbb{R}$ and for all $\lambda > \lambda_0(s)$. Exactly like for the case of $\alpha = 1$, setting $t = 0$ and applying Fatou's lemma ensures that $\liminf_{\eta \downarrow 0} \sum_y |G_{E+i\eta}(x, y)|^2 < \infty$ almost surely. We then apply the Simon-Wolff criterion to obtain spectral localization.

- (ii) The polynomial decay of the eigenfunctions follows directly from this bounded density property and the summability bound by applying standard fractional moment arguments [AW15a, Theorem 7.4].
- (iii) The dynamical bounds follow from bounding the dynamic evolution using the eigenfunction correlator $Q(x, 0, \Sigma)$. We note that:

$$\sup_{t \in \mathbb{R}} |e^{-itH_{\alpha, \omega}}(0, x)|^2 \leq \sup_{F \in C(\mathbb{R}), \|F\| \leq 1} |F(H_{\alpha, \omega})(0, x)|^2 =: Q(x, 0, \Sigma),$$

where Σ denotes the almost sure spectrum of $H_{\alpha, \omega}$. Recall from Chapter 1 that the probability measure μ of the random potential has a bounded density ρ with compact support. Because of this, we can observe that

$$\sup_{v \in \mathbb{R}} (1 + |v|)^s \rho(v) < \infty.$$

The expectation of this correlator is majorized by the integrated Green function moments:

$$\mathbb{E}[Q(x, 0, \Sigma)] \leq C_s \liminf_{\eta \downarrow 0} \int_{\Sigma} \mathbb{E}[|G_{E+i\eta}(x, 0)|^s] dE,$$

for a constant $C_s > 0$ [AW15b, Theorem 7.7]. Using the polynomial decay established in (6.2.2), we obtain:

$$\sum_{x \neq 0} |x|^\beta \mathbb{E}[Q(x, 0, \Sigma)] \leq \frac{C_s K_0 \theta_s}{\lambda^s} \sum_{x \neq 0} |x|^\beta \frac{1}{|x|^{d+2\alpha_s}} < \infty,$$

which converges for all $\beta < s(d + 2\alpha) - d$, or equivalently $\beta < 2\alpha_s$. Since $0 \leq Q(x, 0, \Sigma) \leq 1$, we can conclude that for almost every $\omega \in \Omega$:

$$\sup_{t \in \mathbb{R}} \sum_{x \neq 0} |x|^\beta |e^{-itH_{\alpha, \omega}}(0, x)|^2 \leq \sum_{x \neq 0} |x|^\beta Q_\omega(x, 0)^2 \leq \sum_{x \neq 0} |x|^\beta Q_\omega(x, 0) < \infty,$$

which yields the required dynamical bounds.

□

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